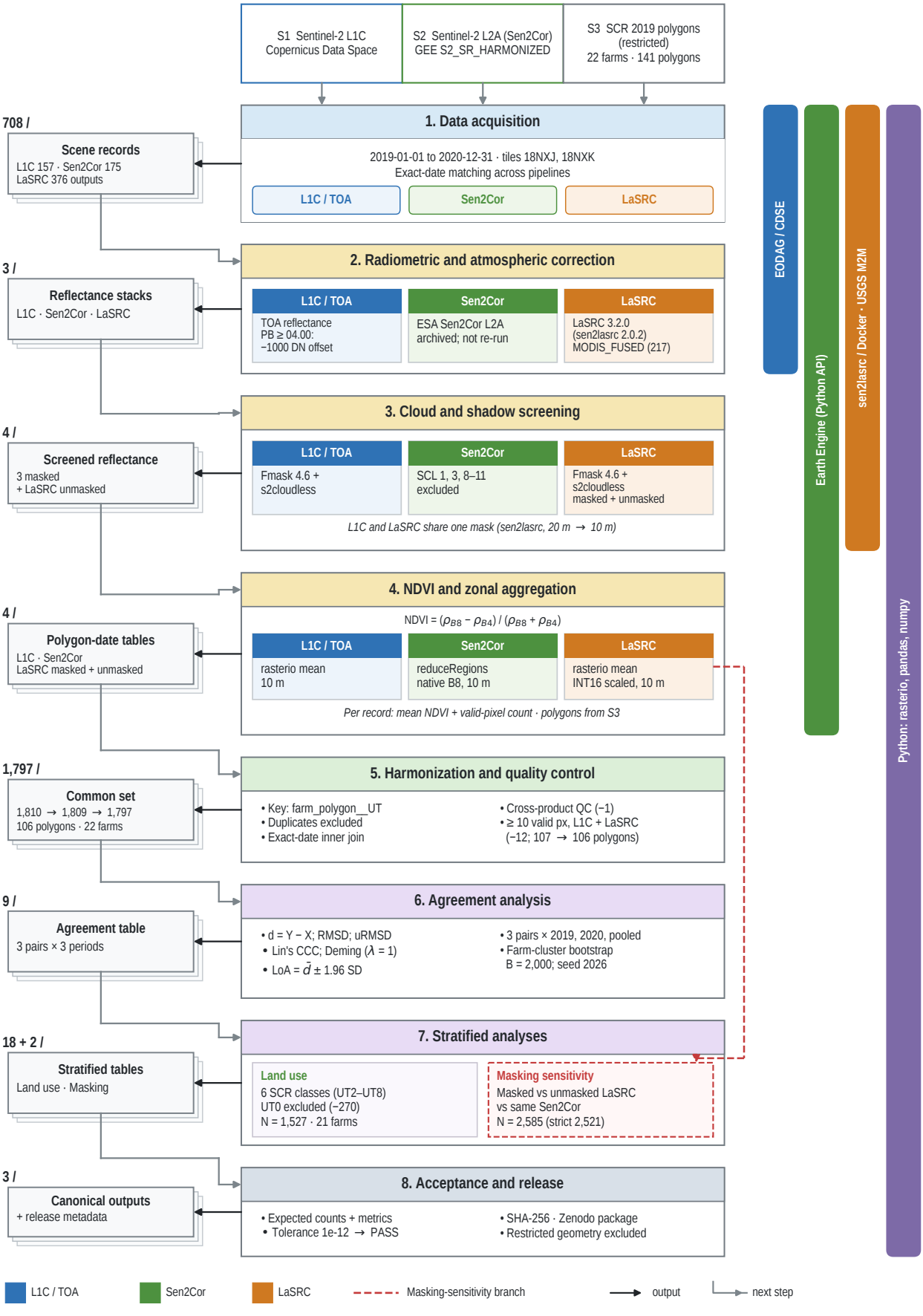

S1 Sentinel-2 L1C
Copernicus Data Space
S2 Sentinel-2 L2A (Sen2Cor)
GEE S2_SR_HARMONIZED
S3 SCR 2019 polygons
(restricted)
22 farms · 141 polygons
1. Data acquisition
2019-01-01 to 2020-12-31 · tiles 18NXJ, 18NXK
Exact-date matching across pipelines
L1C / TOA
Sen2Cor
LaSRC
708 /
Scene records
L1C 157 · Sen2Cor 175
LaSRC 376 outputs
2. Radiometric and atmospheric correction
L1C / TOA
TOA reflectance
PB ≥ 04.00:
−1000 DN offset
Sen2Cor
ESA Sen2Cor L2A
archived; not re-run
LaSRC
LaSRC 3.2.0
(sen2lasrc 2.0.2)
MODIS_FUSED (217)
3 /
Reflectance stacks
L1C · Sen2Cor · LaSRC
3. Cloud and shadow screening
L1C / TOA
Fmask 4.6 +
s2cloudless
Sen2Cor
SCL 1, 3, 8–11
excluded
LaSRC
Fmask 4.6 +
s2cloudless
masked + unmasked
L1C and LaSRC share one mask (sen2lasrc, 20 m → 10 m)
4 /
Screened reflectance
3 masked
+ LaSRC unmasked
4. NDVI and zonal aggregation
NDVI = (ρB8 − ρB4) / (ρB8 + ρB4)
L1C / TOA
rasterio mean
10 m
Sen2Cor
reduceRegions
native B8, 10 m
LaSRC
rasterio mean
INT16 scaled, 10 m
Per record: mean NDVI + valid-pixel count · polygons from S3
4 /
Polygon-date tables
L1C · Sen2Cor
LaSRC masked + unmasked
1,797 /
Common set
1,810 → 1,809 → 1,797
106 polygons · 22 farms
5. Harmonization and quality control
• Key: farm_polygon__UT
• Duplicates excluded
• Exact-date inner join
• Cross-product QC (−1)
• ≥ 10 valid px, L1C + LaSRC
(−12; 107 → 106 polygons)
9 /
Agreement table
3 pairs × 3 periods
6. Agreement analysis
• d = Y − X; RMSD; uRMSD
• Lin's CCC; Deming (λ = 1)
• LoA = d̄ ± 1.96 SD
• 3 pairs × 2019, 2020, pooled
• Farm-cluster bootstrap
B = 2,000; seed 2026
18 + 2 /
Stratified tables
Land use · Masking
7. Stratified analyses
Land use
6 SCR classes (UT2–UT8)
UT0 excluded (−270)
N = 1,527 · 21 farms
Masking sensitivity
Masked vs unmasked LaSRC
vs same Sen2Cor
N = 2,585 (strict 2,521)
3 /
Canonical outputs
+ release metadata
8. Acceptance and release
• Expected counts + metrics
• Tolerance 1e-12 → PASS
• SHA-256 · Zenodo package
• Restricted geometry excluded
EODAG / CDSE
Earth Engine (Python API)
sen2lasrc / Docker · USGS M2M
Python: rasterio, pandas, numpy
L1C / TOA
Sen2Cor
LaSRC
Masking-sensitivity branch
output
next step